Data Request: Target Trial Emulation of ARUBA

Requested by: Emily Molins (DPhil) · **Audience:** Truveta Data Engineering **Purpose:** Define the patient cohort and data elements needed to emulate the ARUBA randomized trial in real-world EHR data.

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0. Plain-language overview (start here)

This section assumes no neurovascular background — it explains what we're studying and what we need.

What is a brain AVM? A brain **arteriovenous malformation (AVM)** is a tangle of abnormal blood vessels where arteries connect directly to veins, bypassing the normal capillaries. It can **rupture and cause a brain bleed (a haemorrhagic stroke)**. An **"unruptured" AVM** is one found *before* any bleed — often incidentally, or after a seizure or headache.

The clinical dilemma. For an *unruptured* AVM, doctors face a hard choice: **treat it** (via surgery, catheter-based "embolisation," or focused radiation — each of which carries its own risk of causing a stroke), or **leave it alone and monitor** ("conservative management") and accept the future rupture risk. There is no clear answer, and it is one of the most debated decisions in neurovascular medicine.

What ARUBA was. ARUBA was the single major **randomized controlled trial (RCT)** that tried to settle this [1,3]. It controversially concluded that *monitoring* led to better outcomes than *treating* — but it was **stopped early**, enrolled few patients, and used treatment approaches many surgeons considered non-standard, so the result is heavily disputed [1,2,5,13].

What we want to do. We want to re-examine ARUBA using a **target trial emulation (TTE)** — a method that uses large real-world patient records to *mimic* a randomized trial when running a new RCT isn't feasible [11,12]. Done well, this lets us re-test "treat vs. monitor" at **far larger scale, with modern practice and longer follow-up** than ARUBA had.

What we need from Truveta (the ask in one line). A cohort of patients with an **unruptured brain AVM**, with enough clinical detail — *especially a measure of how severe/complex the AVM is* — and **long-term follow-up** to compare those who were treated vs. monitored.

The one catch to flag up front. The standard measure of AVM severity (the **Spetzler–Martin grade**) is read off **angiogram/MRI imaging and radiology reports** — information that usually lives in *clinical notes*, not in standard billing/diagnosis codes. So the central question for this request is: **what note-derived "concepts" can Truveta**

provide to capture AVM severity? Without some severity signal, the comparison is confounded (see §2, §3).

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1. Background & rationale (with citations)

1a. What ARUBA was. Multicentre (39 sites), non-blinded RCT; 226 adults (≥18) with an **unruptured** brain AVM, randomised 1:1 to **intervention** (neurosurgery, embolisation, and/or stereotactic radiosurgery — alone or combined) vs. **medical management alone** [1,3]. Primary outcome: composite of **death (any cause) or symptomatic stroke** (haemorrhage or infarction confirmed on imaging) [1,3]. Planned follow-up **5–10 years** [3]; **halted early** (mean ~33 months) with the intervention arm doing worse (≈30.7% vs 10.1%) [1]. Longer follow-up (2020) still favoured medical management [2].

1b. Why ARUBA is contested. - **Non-representative treatment:** few patients underwent microsurgical resection — the recognized gold standard was under-used — and the intervention arm was heterogeneous and non-standardised [5,13]. - **Selection bias & low enrolment:** thousands screened, only 226 randomised → poor generalizability [5,13]. - **Short follow-up** for a disease with *lifelong, cumulative* rupture risk [7,13]. - Practice shifted toward conservative management afterward, and national-sample data show an increased *proportion* of ruptured AVM presentations (≈17% → 23%) and worse in-hospital outcomes in the post-ARUBA era [8].

1c. Previous attempts to re-examine it — and their shortcomings. - **Scottish population cohort (SIVMS):** observational comparison that *supported* conservative management — but region-specific and observational [4]. - **Single-centre ARUBA-eligible cohorts:** suggest good outcomes with modern microsurgery — but small and selection-biased [5]. - **Systematic review / meta-analysis of interventional outcomes in ARUBA-eligible patients (2021):** pooled observational data, not a formal causal emulation [6]. - **30-year single-centre conservative series:** rare long-term data on *monitored* AVMs — still single-centre [7]. - **Common shortcomings:** small / single-centre, selection bias, no formal target-trial design, and scarce long-term data on conservatively managed AVMs.

1d. Why a target trial emulation adds value. A **large-scale, multi-site EHR emulation** with an explicit protocol, modern real-world practice, longer follow-up, and formal causal methods can test *treat vs. monitor* at a scale and generalizability ARUBA and its re-analyses could not [11,12]. It directly informs a high-stakes, still-unresolved decision.

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2. Eligibility & target-trial protocol

We follow the standard 7-component target-trial protocol [11]: eligibility, treatment strategies, assignment, time zero + follow-up, outcome, causal contrast, analysis.

2a. Eligibility (mirroring ARUBA) [1,3] - **Include:** adults ≥18; **incident, unruptured** brain AVM (ICD-10 **Q28.2**); AVM considered treatable; baseline functional status **mRS < 2**. - **Exclude:** any **prior intracranial haemorrhage** (ICD-10 I60–I62) before the index date;

prior AVM treatment (embolisation, resection, or radiosurgery before index); baseline mRS ≥ 2.

2b. Treatment strategies (the two arms) - **Intervention:** surgery / embolisation / radiosurgery (alone or combined) started within a defined **grace period** after eligibility. - **Conservative:** no intervention (symptom management only).

2c. Time zero & assignment - **Time zero** = date of first AVM diagnosis / eligibility. Handle the grace period and immortal-time bias via **clone-censor-weight** or sequential trial emulation [11].

2d. Outcome - Primary composite: **all-cause death** + **symptomatic stroke** (intracranial haemorrhage / infarction, imaging-confirmed) [1]. Requires **mortality linkage** + haemorrhage/infarction codes (I60–I63). - Secondary (if available): functional decline (mRS change), seizure control, procedural complications.

2e. Longitudinality - Need **long, continuous follow-up** (5+ years, ideally 10) with low loss to follow-up, to capture cumulative rupture risk [7,13]. → *Ask Truveta: median continuous enrolment / follow-up for this cohort.*

2f. Scale - ARUBA randomised 226 from thousands screened [1]; an EHR cohort should yield far more. → *Ask Truveta: count of Q28.2 patients, then the unruptured + treatment-naïve + mRS<2 subset.*

2g. Confounding control (why §3 matters) - The central threat is **confounding by indication**: larger, more eloquent AVMs are treated differently, and AVM severity independently drives outcomes [4,13]. Adjusting for **severity** is therefore essential — this is the crux of the data request. - Baseline covariates to request: age, sex, **AVM severity (§3)**, presentation (incidental vs. seizure vs. headache), comorbidities, calendar year, site/region.

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3. Concept codes — capturing AVM severity

The problem for Truveta. The standard severity score — the **Spetzler–Martin (SM) grade** — is derived from **angiographic imaging features**, which are described in radiology reports/notes rather than standard codes [9,10]. Because Truveta does not hold the images, severity must come from **note-derived "concepts."**

3a. Spetzler–Martin components to request as concepts (from reports) [9] - **Nidus size:** <3 cm (1 pt) · 3–6 cm (2 pts) · >6 cm (3 pts) - **Eloquence** of location (sensorimotor, language, visual cortex, thalamus, internal capsule, brainstem, deep cerebellum): 0/1 - **Deep venous drainage:** 0/1 - Derived **SM grade (I–V)** or **Spetzler–Ponce class (A/B/C)** if extractable - Supplemental (Lawton–Young) factors where available: nidus **compactness/diffuseness** [10]

3b. Codeable proxies already in structured data - **Diagnosis:** cerebral AVM — ICD-10 **Q28.2** - **Rupture / haemorrhage (exclusion + outcome):** SAH **I60**, intracerebral haemorrhage **I61**, other intracranial haemorrhage **I62**, cerebral infarction **I63** - **Presentation:** seizure/epilepsy **G40**; headache - **Treatments (define arms; wash out prior treatment):** microsurgical resection (CPT 61680–61692, “surgery of intracranial AVM”), transcatheter embolisation (CPT 61624 and related S&I codes), stereotactic radiosurgery / Gamma Knife (CPT 61796–61800; 77371–77373) — *confirm exact code sets with the team* - **Functional status:** modified Rankin Scale (**mRS**) — for the mRS<2 entry criterion and outcomes; often not coded → *ask whether it exists as a concept* - **Adjuncts:** NIHSS, hydrocephalus, associated aneurysm

3c. Explicit questions for Truveta 1. Which SM components (size, eloquence, venous drainage) are available as extracted concepts, and at what coverage? 2. Is an SM grade or mRS captured as a concept anywhere? 3. What is the cost/effort per concept, and can radiology-report concepts be provisioned **before the December 2026 access cutoff**?

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4. Summary of the request

We are asking Truveta for a **cohort and data extract** to support the ARUBA emulation:

1. **Cohort:** adults ≥18 with unruptured brain AVM (Q28.2), treatment-naïve, mRS<2 at baseline — with counts at each filter step.
2. **Arms:** treatment procedures (surgery / embolisation / radiosurgery) to define intervention vs. conservative.
3. **Outcomes:** all-cause mortality (linkage) + haemorrhage/infarction (I60–I63).
4. **Follow-up:** longest available continuous enrolment (5–10 yr target).
5. **Severity concepts (critical):** SM components / grade and mRS from radiology-report concepts — availability, coverage, cost, timeline.
6. **Covariates:** age, sex, presentation, comorbidities, calendar year, site.

Success criterion: whether enough **severity signal** can be obtained from concepts to control confounding-by-indication — the factor at the heart of the ARUBA debate.

References

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